E-COMMERCE SALES COMPANY

Project objective:

The main objective of this capstone project is to analyse the e-commerce data of a company by performing some mandatory tasks includes data cleaning, data exploration, data modelling and creating pivot tables and making dashboard visualization using Microsoft Excel.

Data Description:

The given data set consist of four sheets Customer_dim, Product_dim, Store_dim, Sales_fact where Sales_fact table linked with other three tables by Customer_id, Product_id, Store_id respectively. The dataset mainly focused on sales transaction ha for the period of oct-2023 to oct-2025

Table Description:

Customer_dim:

It consists of unique Customer_id and their respective age, name, gender, city, state, country and their loyalty level. There was a three blank Values at Loyalty level column and I replaced it with a mode=" Platinum".

Product_dim:

It consists of a unique product_id with respective product name, category, sub-category, Brand, Stock and Cost. There were a few blank rows in Stock column, so it's a numerical column I used an average function to fill the values.

Store_dim:

It consists of a unique Store_id and respective store name, region, city and store type.

Sales_fact:

It contains unique sales_id for each transaction. It has the records of transaction happened for a period of oct,2023 to oct, 2025.It has multiples of Customer_id, Product_id, Store_id and the quantity, cost, discount and total cost of each transaction. To put a value in a blank cell I used a mathematical formula separately for each column.

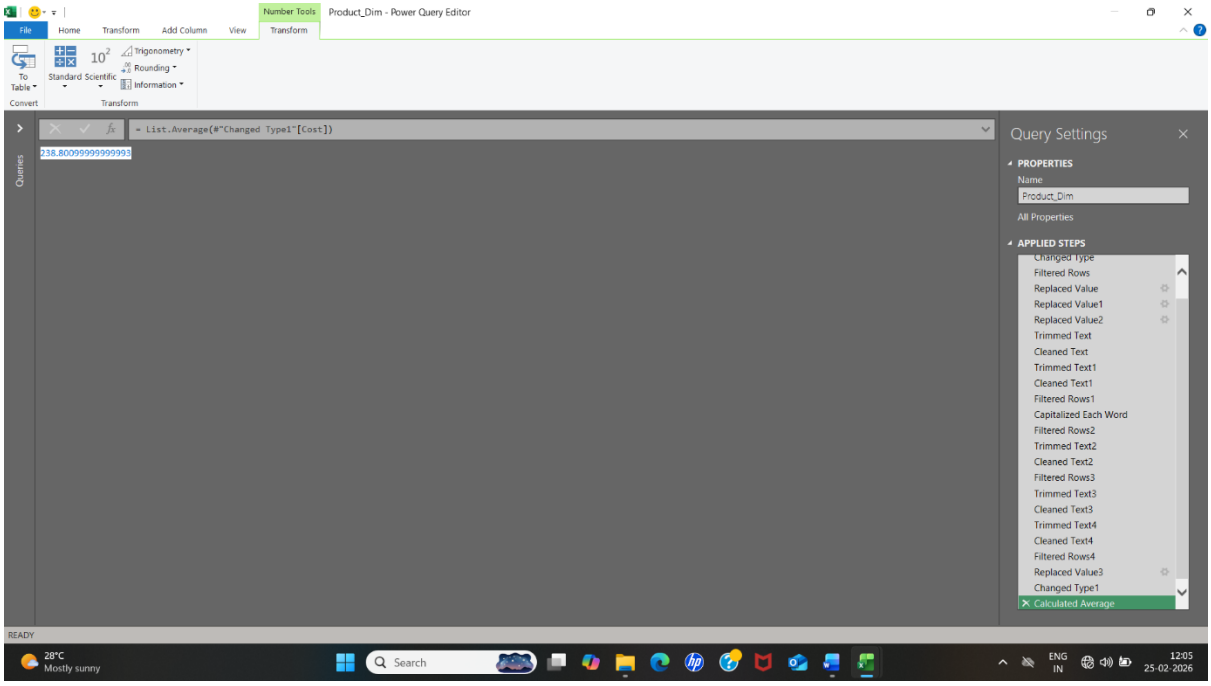
Data Cleaning and Formulae used:

Product_dim:

Using Power Query 1. Find and Replace

2. Clean and Trim and find duplicates
3. Capitalize Each Word
4. Used statistics method to find average to fulfil the empty cells.

Average=238.8



Using Formulae 5. In Stock column, replace empty values with average using formula

=INT(IF(ISBLANK([@Stock]),AVERAGE([Stock]),[@Stock]))

Store_dim:

Using Power Query 1. Find and Replace

2. Clean and Trim and find duplicates

3. Capitalize Each Word

Customer_dim:

Using Power Query 1. Find and Replace

2. Clean and Trim and find duplicates

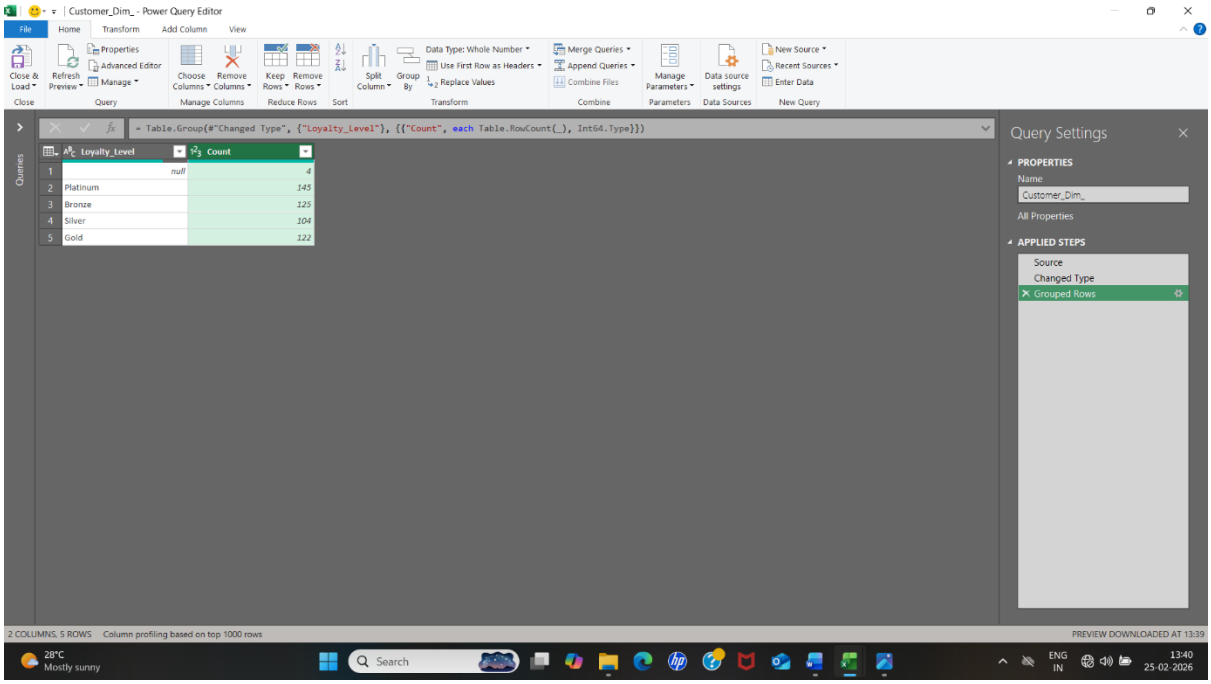
3. Capitalize Each Word

4.In loyalty_level column to find a mode,

I used Group By and replaced an empty cell with Platinum.

Platinum – 145 Silver - 104

Bronze - 125 Gold - 122



Sales_fact:

1. Clean and Trim
2. Replacing empty cells using following formulae

Quantity

=IF(ISBLANK([@Quantity]),[@[Total_Amount]]/([@[Unit_Price]]*(1-[@Discount])),[@Quantity])

Unit Prize

=IF(ISBLANK([@[Unit_Price]]),[@[Total_Amount]]/[@Quantity]*(1-[@Discount]),[@[Unit_Price]])

Discount

=IF(ISBLANK([@Discount]),1-([@[Total_Amount]]/([@Quantity]*[@[Unit_Price]])),[@Discount])

Total Amount

= IF(ISBLANK(Sales_Fact[@[Total_Amount]]),Sales_Fact[@Quantity]*Sales_Fact[@[Unit_Price]]*(1-Sales_Fact[@Discount]),Sales_Fact[@[Total_Amount]])

Final_sales:

The Final_Sales table was created by duplicating the Sales_Fact table. Using the VLOOKUP function, Category and Original Cost were extracted from the Product_Dim table, and Store Type was retrieved from the Store_Dim table. Based on this data, Total Cost and Final Profit columns were calculated to analyse the profit generated.

Category: VLOOKUP([@[Product_ID]],Product_Dim_1,3)

Store type: VLOOKUP([@[Store_ID]],Store_Dim_1,5)

Org Cost: VLOOKUP([@[Product_ID]],Product_Dim_1,7)

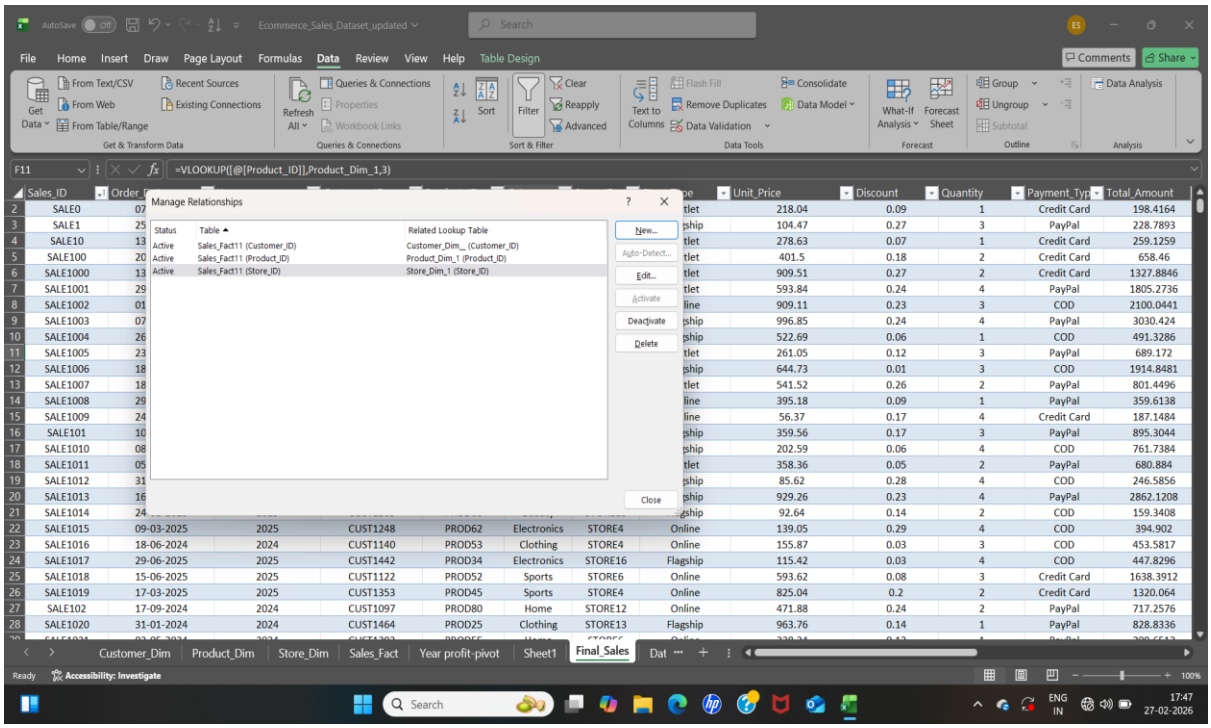
Total Cost: [@[Original Cost]]*[@Quantity]

Profit: [@[Total_Amount]]-[@[Total_cost]]

Descriptive Analysis:

Analysis Tool Pak	
TOTAL PROFIT	
Mean	476.23
Standard Error	17.57
Median	366.00
Mode	#N/A
Standard Deviation	785.86
Sample Variance	617568.52
Kurtosis	0.72
Skewness	0.62
Range	5331.15
Minimum	-1705.51
Maximum	3625.65
Sum	952458.00
Count	2000.00

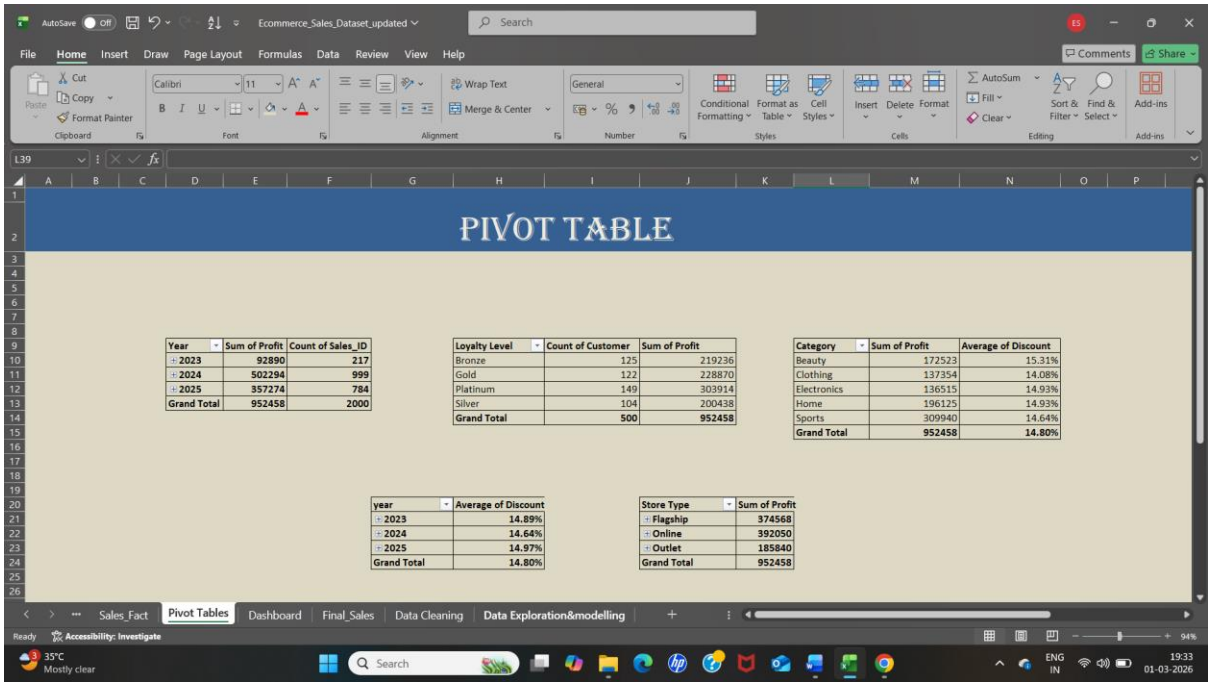
Data Modelling:



Pivot Table:

There are totally a six Pivot Tables

1. Year vs Sum of profit and Count of Sales
2. Payment Type vs Gender
3. Loyalty Level vs Count of Customers and Sum of Profit
4. Year vs Average discount
5. Store Type vs Sum of Profit



Year	Sum of Profit	Count of Sales_ID
2023	92890	217
2024	502294	999
2025	357274	784
Grand Total	952458	2000

Loyalty Level	Count of Customer	Sum of Profit
Bronze	125	219236
Gold	122	228870
Platinum	149	303914
Silver	104	200438
Grand Total	500	952458

Category	Sum of Profit	Average of Discount
Beauty	172523	15.31%
Clothing	137354	14.08%
Electronics	136515	14.93%
Home	196125	14.93%
Sports	309940	14.64%
Grand Total	952458	14.80%

year	Average of Discount
2023	14.89%
2024	14.64%
2025	14.97%
Grand Total	14.80%

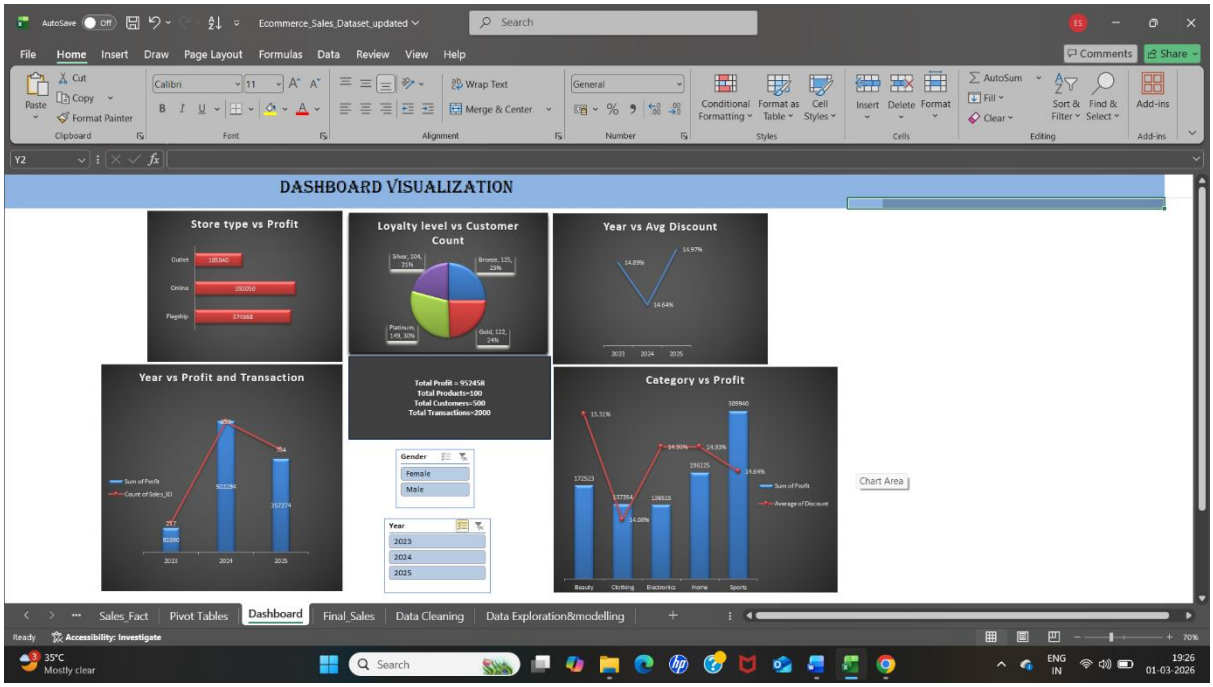
Store Type	Sum of Profit
Flagship	374568
Online	392050
Outlet	185840
Grand Total	952458

Data Visualization:

In this project, I used Microsoft Excel to create charts and an interactive dashboard to understand the sales data clearly.

I used different types of charts like column charts to compare categories and store types, and line charts to see the sales trend by year. I also created KPI cards to show important numbers such as Total Profit, Total Transactions, Total Customers, and Total Products.

I added slicers for Year and Gender so that when a user selects a value, all the charts update automatically. This makes the dashboard interactive and easy to analyse.



Insights about this dataset:

Year vs Profit and Transaction:

1. The data for 2023 includes only October, November, and December, so it does not represent the full year and cannot be fully compared with other years.

2. The data for 2024 includes the complete 12 months, so it shows the highest sales, transactions, and profit among the three years.

3. The data for 2025 includes only the first nine months. Compared to 2024, 2025 has fewer transactions and lower profit for this period.

4. The company should focus on improving promotions and marketing strategies to increase product purchases, especially during major festival seasons.

5. Christmas is one of the biggest festivals in the USA, so special offers and discounts during this time can help boost sales and improve overall performance.

6. Since each year does not have equal data coverage, direct yearly comparison may not give an accurate conclusion. A quarterly comparison would provide better insights.

Loyalty Level vs Customer count:

1. The number of customers in each loyalty level remains mostly consistent across all three years.

2. Platinum customers are higher compared to other loyalty categories.

3. This shows strong retention of high-value customers.

4. The company can focus on upgrading more customers to higher loyalty levels to increase overall revenue.

Year vs Average Discount:

1. The average discount decreased from 14.89% in 2023 to 14.64% in 2024.

2. This reduction in discount percentage in 2024 was accompanied by higher transactions, sales, and overall profit.

3. In 2025, the average discount increased to 14.97%.

4. However, compared to the first three quarters of 2024, 2025 recorded lower transactions and profit despite offering a higher discount rate.

5. This indicates that the increase in discount percentage did not effectively improve revenue performance in 2025.

6.The company should reassess and optimize its discount strategy to enhance profitability and achieve sustainable revenue growth.

Category vs Profit:

2023

1. Sports generated the highest profit of 25,767 with the lowest discount rate of 13.49%.

2. Beauty recorded the third-highest profit (16,716) despite offering the highest discount of 16.61%.

3. Clothing had comparatively lower profit performance.

2024

1. Sports again led with the highest profit of 166,506, offering a relatively lower discount of 14.47%.

2. Electronics had the highest discount rate of 15.25%, yet recorded the lowest profit of 53,960.

3. Clothing remained one of the lower-performing categories.

2025

1. Sports continued to dominate with the highest profit of 117,667, while the discount increased to 15.18%.

2. Clothing offered the lowest discount (14.17%) and recorded the lowest profit (51,149).

3. Electronics improved significantly, moving to second position after Sports.

Analysis:

1. Sports, Home, and Beauty consistently remained top-performing categories in 2023 and 2024.

2. In 2025, Electronics improved, but Clothing still gave the lowest profit.

3. Giving higher discounts did not always increase profit (example: Electronics in 2024).

Recommendation:

1. Sports is performing well even with lower discounts, so no need to give high discounts.

2. Beauty and Home discounts can be slightly reduced to increase profit.

3. Clothing needs better offers or marketing because low discount alone is not improving sales.

Store Type vs Profit:

1. The company has a total of 20 stores: 8 Flagship, 8 Online, and 4 Outlet stores.

2. Flagship and Online stores generate higher profit compared to Outlet stores.

3. In 2024, all store types achieved their highest profit.

4. In 2025, profit decreased compared to 2024, but Flagship and Online still performed better than Outlet.

5. Since Outlet has only 4 stores and generates the lowest profit, the company should improve strategies for Outlet stores to increase performance.

Conclusion:

Overall, the business performed very well in 2024, with the highest profit and number of transactions across all categories and store types. The Sports category gave the highest profit, and Flagship and Online stores were the main contributors to revenue. In 2025, profit and transactions slightly decreased compared to 2024. This may be because the data is incomplete and the discount percentage was increased. Higher discounts did not always increase revenue, so the company should manage discounts carefully. To improve performance, the company should continue focusing on strong categories, improve Outlet store sales, and use special promotions during big festivals like Christmas to increase profit.